Literature Review

No.	Title of the Paper	Authors	Journal Name & Year Published	Methodology Adopted	Key Findings	Gaps
1	Cost-Optimised Machine Learning Model Comparison for Predictive Maintenance	Yating Yang, Muhammad Zahid Iqbal	Electronics, 2025	Compared SVM, Random Forest, Decision Tree, KNN, MLP, XGBoost and LightGBM on sliding-window time-series features. Tested downsampling, SMOTETomek and class weighting to handle class imbalance.	Random Forest and SVM performed strongly with low misclassification cost. A voting ensemble further improved cost efficiency. Downsampling was the most effective imbalance strategy.	Minority-class recall remained low. The study focuses on cost-sensitive model comparison rather than building a simple end-to-end failure prediction system.
2	[Title to be added]	Valeriu Dimidov, Sasan Jafarnejad, Raphaël Frank	2025 IEEE International Conference on Prognostics and Health Management (ICPHM), 2025	Assumed component wear is a monotonically non-decreasing time series. Selected recent observations, converted the time series into tabular features, and trained ML models through an AutoML framework. SHAP was used for explainability.	Reduced predictive-maintenance costs compared with existing approaches, while AutoML simplified the modelling process.	The simplified representation uses mainly recent observations and may lose temporal information. Advanced temporal or deep-learning models could capture degradation patterns better.
3	Enhancing Predictive Maintenance for Heavy Vehicles: A Multi-Stage Feature Selection and Preprocessing Framework on the Scania Dataset	Stefania Ferrisi, Romina Conte, Rosita Guido, Giuseppina Ambrogio	Procedia Computer Science, 2026	Proposed a multi-stage preprocessing and feature-selection framework to handle missing values, anonymized features and class imbalance. Compared several ML models.	Improved robustness and transparency, and showed potential for more accurate failure prediction and lower maintenance and replacement costs.	Focuses mainly on preprocessing and feature selection. More ML models, temporal feature engineering and generalization settings remain to be explored.
4	Two-Stage LightGBM Framework for Cost-Sensitive Prediction of Impending Failures of Component X in Scania Trucks	Si-Woo Kim, Yong Soo Kim	Computers, Materials & Continua, 2026	Converted recent operational records into last-k statistical and trend features. Used a two-stage LightGBM model: Stage A detects normal vs. faulty condition, and Stage B predicts failure severity with cost-sensitive classification.	Achieved lower misclassification cost than XGBoost and Bi-LSTM while keeping high recall for safety-critical failures. Handles imbalance, noisy data and irregular sampling.	Optimized only for LightGBM and cost-sensitive severity classification. Future work: better temporal modelling, data augmentation, intermediate-failure detection and adaptive thresholds.

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5	SCANIA Component X Dataset: A Real-World Multivariate Time Series Dataset for Predictive Maintenance	Kharazian, Lindgren, Magnússon, Steinert, Andersson Reyna	Scientific Data, 2025	Documented the construction of a real-world heavy-vehicle dataset with vehicle records and multivariate sensor signals.	Provides a realistic benchmark for imbalanced failure prediction.	Vehicle-level validation and temporal drift need further study.
6	Enhancing Predictive Maintenance in Automotive Industry: Addressing Class Imbalance Using Advanced Machine Learning Techniques	Mahale, Kolhar, More	Discover Applied Sciences, 2025	Applied resampling, class-imbalance analysis and ML/DL model comparisons to rare failure events in automotive data.	Precision and recall are more informative than accuracy alone for imbalanced data.	Robust imbalance-handling strategies need evaluation on external fleets.
7	Predictive Maintenance in Industry 4.0: A Survey of Planning Models and Machine Learning Techniques	Hector, Panjanathan	PeerJ Computer Science, 2024	Surveyed supervised, unsupervised and semi-supervised methods for predictive maintenance planning.	Algorithm choice depends on the data type, objective and deployment context.	Model selection should be linked more closely with actual maintenance decisions.